Nuclear Physics Angular Momentum Coupling Guide

This document provides the theoretical background and practical usage guide for the `angular_momentum_coupling.py` utility. This tool calculates allowed final nuclear states based on quantum mechanical selection rules.

1. Theoretical Basis

The code implements the **Channel Spin Coupling Scheme**, which is a standard method for determining angular momentum conservation in nuclear reactions.

Notation

Symbol	Meaning
J_{target}	Spin of target nucleus
π_{target}	Parity of target nucleus
$s_{particle}$	Spin of transferred particle (Python input)
$\pi_{particle}$	Parity of transferred particle (Python input)
$T_{particle}$	Isospin of transferred particle
ℓ	Relative orbital angular momentum between the target and the particle
S	Channel Spin = $J_{target} + s_{particle}$
J_{final}	Spin of the populated final nuclear state
π_{final}	Parity of the populated final nuclear state

Conservation Laws

In any nuclear reaction $Target + Particle \to Final_State$, the following quantities are conserved:

1. **Total Angular Momentum** ($\vec{J}$):

   $$\vec{J}_{final} = \vec{J}_{target} + \vec{s}_{particle} + \vec{\ell}$$

2. **Parity** (π):

   $$\pi_{final} = \pi_{target} \times \pi_{particle} \times (-1)^{\ell}$$

Coupling Scheme Used in Code

The code performs the vector addition in two steps (Channel Spin representation):

1. **Calculate Channel Spin** ($\vec{S}$): Couples the target spin and particle spin.

   $$\vec{S} = \vec{J}_{target} + \vec{s}_{particle}$$

Possible values: $|J_{target} - s_{particle}| \le S \le J_{target} + s_{particle}$

2. **Calculate Final Spin** ($\vec{J}_{final}$): Couples the channel spin with the orbital angular momentum.

   $$\vec{J}_{final} = \vec{\ell} + \vec{S}$$

   Possible values: $|\ell - S| \le J_{final} \le \ell + S$

Important: The "Particle" input to this tool represents the **intrinsic properties of the transferred particle**: spin $s_{particle}$, parity $\pi_{particle}$, and isospin $T_{particle}$.

2. Application to Different Reaction Types

To use the code correctly, you must identify the correct "Particle" input based on the reaction mechanism.

A. Resonant Capture Reactions

Examples: (p, γ), (n, γ)

In these reactions, a projectile fuses with the target to form a compound nucleus (resonance).

- **Particle Input**: `1/2+`

B. Single-Nucleon Transfer Reactions

Examples: (d, p), (p, d), $(^3\text{He}, d)$, $(d, ^3\text{He})$, (n, d), (d, n), (d, t), (t, d), (α, t), (t, α), $(^3\text{He}, \alpha)$, $(\alpha, ^3\text{He})$

In these reactions, a single nucleon is transferred between the projectile and the target. You need to input the properties of the **transferred nucleon**.

- **Particle Input**: `1/2+`

C. Two-Nucleon Transfer Reactions

In these reactions, a pair of nucleons is transferred. The total spin of this pair depends on the coupling of the two nucleons.

1. Identical Nucleon Transfer (2n or 2p)

Reactions: (p, t), (t, p) (2n transfer); $(^3\text{He}, n)$ (2p transfer)

For the transfer of two identical nucleons (2n or 2p) in the same shell model orbit, the **Pauli Exclusion Principle** dictates that their total wavefunction must be antisymmetric.

- If they are in a relative s-state (spatially symmetric, $L=0$), their **spin wavefunction must be antisymmetric**.
- An antiparallel spin state for two fermions corresponds to a **$S=0$** (spin singlet).
- **NDS Policy**: For (p, t), (t, p), and $(^3\text{He}, n)$ reactions, it is standard to assume the transferred pair is in a relative s state ($S=0$).
- **Particle Input**: `0+`

2. Neutron-Proton Transfer (1n1p)

Reactions: (α, d), (d, α), $(^3\text{He}, p)$, $(p, ^3\text{He})$

For the transfer of a neutron-proton pair, the selection rules depend on the projectile/ejectile properties.

- **(α, d) and (d, α)**:
    - Both d and α have $T=0$. Thus, only **$T=0$** transfer is allowed.
    - Antisymmetry requires $S+T$ to be odd (for relative $L=0$). Since $T=0$, the transfer is restricted to **$S=1$** (spin triplet).
    - **Particle Input**: `1+`

- **$(^3\text{He}, p)$ and $(p, ^3\text{He})$**:
    - Both particles have $S=1/2, T=1/2$.
    - Allowed transfers are **$(S=0, T=1)$** and **$(S=1, T=0)$**.
    - **Particle Input**:
        - For $S=0$ component: `0+`
        - For $S=1$ component: `1+`
    - *Note: Both components can contribute.*

D. Cluster Transfer Reactions

Examples: $(^6\text{Li}, d)$, $(^7\text{Li}, t)$

An alpha particle is transferred.

- **Particle Input**: `0+`

E. Inelastic Scattering

Examples: (α, α'), (p, p')

- **(α, α')**: The alpha particle has spin 0. Since the projectile spin cannot flip, the angular momentum transfer is purely orbital ($\vec{\ell}$). This selectively excites **Natural Parity** resonances ($\pi = (-1)^\ell$) in the compound nucleus.

- **Particle Input**: `0+`

- **(p, p')**: The proton ($s=1/2$) can undergo spin-flip. It allows both Isoscalar ($T=0$) and Isovector ($T=1$) transitions. It populates both Natural and Unnatural parity resonances.

- **Particle Input**: For the $S=0$ component use `0+`; for the $S=1$ component use `1+`.

F. Charge Exchange Reactions

Examples: (p, n), $(^3\text{He}, t)$, $(^6\text{Li}, ^6\text{He})$

These reactions exchange a nucleon type (isospin flip), strictly requiring an isospin transfer of $\Delta T = 1$.

- **Fermi Transitions** ($\Delta S=0, \Delta T=1$): Excites the Isobaric Analog State (IAS).

    - **Particle Input**: `0+`
- **Gamow-Teller Transitions** ($\Delta S=1, \Delta T=1$): Excites 1^+ states from 0^+ targets.
    - **Particle Input**: `1+`

Specific Selectivity:

- **(p, n) and $(^3\text{He}, t)$**: Mixed probes. Run with both `0+` and `1+`.
- **$(^6\text{Li}, ^6\text{He})$**: Pure Gamow-Teller probe due to the $1^+ \to 0^+$ projectile transition. Fermi transitions are forbidden.
    - **Particle Input**: `1+`

3. Interpreting the Output

When you run the code:

```
python .github/scripts/angular_momentum_coupling.py 3/2- 1/2+
```

You receive output like:

```
Target: J=3/2 π=- | Particle: s=1/2 π=+
Channel Spins S: 1, 2: from |3/2 - 1/2| to 3/2 + 1/2
------------------------------------------------------------
L   Wave  Parity   S     Final Jπ
------------------------------------------------------------
0   s     -        1     1-
0   s     -        2     2-
------------------------------------------------------------
1   p     +        1     0+, 1+, 2+
1   p     +        2     1+, 2+, 3+
------------------------------------------------------------
```

- **L (Wave)**: The orbital angular momentum (ℓ) between target and particle, ranging from 0 to 4 in the output (s, p, d, f, g waves).
- **Parity**: The parity of the final state, determined by $\pi_{final} = \pi_{target} \times \pi_{particle} \times (-1)^{\ell}$.
- **S (Channel Spin)**: The intermediate coupling $S = \vec{J}_{target} + \vec{s}_{particle}$ before adding orbital angular momentum.
    - For $3/2^-$ target and $1/2^+$ particle: S ranges from $|3/2 - 1/2| = 1$ to $3/2 + 1/2 = 2$.
- **Final Jπ**: All allowed total angular momentum and parity combinations for each (L, S) pair, calculated from $|\ell - S| \le J_{final} \le \ell + S$.

Example Analysis: $^{59}\text{Cu}(p, \gamma)^{60}\text{Zn}$

- **Target**: ^{59}Cu ($3/2^-$)
- **Particle**: Proton ($1/2^+$)

- **Channel Spins**: $S = 1, 2$ (from $|3/2 - 1/2|$ to $3/2 + 1/2$)
- **Results**:
  - **s-wave capture** ($L=0$): Parity = $(-) \times (+) \times (-1)^0 = -$
    - $S=1$: $J^\pi = 1^-$
    - $S=2$: $J^\pi = 2^-$
  - **p-wave capture** ($L=1$): Parity = $(-) \times (+) \times (-1)^1 = +$
    - $S=1$: $J^\pi = 0^+, 1^+, 2^+$ (from $|1-1|$ to $1+1$)
    - $S=2$: $J^\pi = 1^+, 2^+, 3^+$ (from $|2-1|$ to $2+1$)
  - **d-wave capture** ($L=2$): Parity = $-$
    - $S=1$: $J^\pi = 1^-, 2^-, 3^-$
    - $S=2$: $J^\pi = 0^-, 1^-, 2^-, 3^-, 4^-$

4. Isospin Coupling

For transfer reactions, isospin (T) provides additional selection rules.

- **Single Nucleon Transfer**: Transferring one nucleon (n or p) has $T_{particle} = 1/2$.
- **Identical Nucleon Pair (2n/2p)**: In a relative $L=0$ state, the Pauli Principle requires $S=0$ (spin singlet) and $T=1$ (isospin triplet). Thus $T_{particle} = 1$.
- **Neutron-Proton Pair (np)**:
  - Deuteron-like ($S=1, T=0$): $T_{particle} = 0$
  - Singlet ($S=0, T=1$): $T_{particle} = 1$

Selection Rule: The final isospin is determined by vector addition:

$$\vec{T}_{final} = \vec{T}_{target} + \vec{T}_{particle}$$

Note: This tool calculates angular momentum (J^π) only. Isospin selection rules must be applied separately.

5. Summary of Transferred Particle Properties

Reaction Mechanism	Common Examples	Physical Process	$s_{particle}$	$T_{particle}$
Radiative Capture	$(p,\gamma), (n,\gamma)$	Projectile capture	$1/2$	$1/2$
	(α,γ)	Projectile capture	0	0
Inelastic Scattering	(α,α')	Excitation (Natural Parity)	0	0
	(p,p')	Excitation ($S=0$ and $S=1$)	0 and 1	0 and 1
One Neutron Transfer	$(d,p), (p,d), (t,d), (d,t), (\alpha,^3\text{He}), (^3\text{He},\alpha)$	Transfer of n	$1/2$	$1/2$

Reaction Mechanism	Common Examples	Physical Process	$s_{particle}$	$T_{particle}$
One Proton Transfer	$(d,n), (n,d), (^3\text{He},d), (d,^3\text{He}), (\alpha,t), (t,\alpha)$	Transfer of p	$1/2$	$1/2$
Two-Nucleon Transfer	$(p,t), (t,p), (^3\text{He},n)$	Transfer of $2n$ or $2p$ ($L_{rel}=0$)	0	1
	$(\alpha,d), (d,\alpha)$	Transfer of np (Deuteron-like)	1	0
	$(^3\text{He},p), (p,^3\text{He})$	Transfer of np (Mixed)	0 and 1	1 ($S=0$) / 0 ($S=1$)
Cluster Transfer	$(^6\text{Li},d)$	Transfer of α-cluster	0	0
Charge Exchange	$(p,n), (n,p)$	Fermi / Gamow-Teller	0 and 1	1
	$(^3\text{He},t), (t,^3\text{He})$	Fermi / Gamow-Teller	0 and 1	1
	$(^6\text{Li},^6\text{He})$	Pure Gamow-Teller ($1^+ \to 0^+$)	1	1